

Total No. of Questions : 8]

SEAT No. :

PD-4210

[Total No. of Pages : 3

[6403]-3
T.E. (Civil)
DESIGN OF STEEL STRUCTURES
(2019 Pattern) (Semester - V) (301003)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q. 1 or Q. 2, Q. 3 or Q. 4, Q. 5 or Q. 6 and Q. 7 or Q. 8.
- 2) Neat sketches must be drawn wherever necessary
- 3) Figures to the right indicate full marks
- 4) Take $f_y = 250$ and $f_e = 410$ grade of steel
- 5) Take ultimate stress in bolt, $f_{ub} = 400 \text{ N/mm}^2$.
- 6) Assume suitable data, if necessary
- 7) Use of electronic pocket calculator, IS: 800-2007 and steel table are allowed
- 8) Use of cell phone is prohibited in the examination hall.

Q1) a) Define beam-Column with suitable sketches. **[3]**

- b) A column section ISHB 350 @66.12 kg/m carries a factored axial compressive load of 1645 kN and factored B.M. of 90 kN-m. Design a moment resisting base plate. Assume concrete pedestal of M20 grade. **[14]**

OR

Q2) a) Differentiate between slab base and gusseted base. **[3]**

- b) A column having effective length of 3.5 m is subjected to factored axial load of 450 kN and factored moment of 50 kNm. Design the column section. Check for section strength only. **[14]**

Q3) a) Explain in brief web buckling and web crippling with suitable sketches. **[4]**

- b) A simply supported steel joist of 4 m effective span carries a working uniformly distributed load 40 kN/m on entire span and a point load of 50 kN at mid span. The section is laterally supported throughout the span. Design an appropriate section. Apply usual checks for strength along with check for deflection. **[14]**

OR

P.T.O.

- Q4) a)** Explain modes of failure of beam with suitable sketches [4]
- b)** Design a suitable I section for simply supported beam of span 5 m carrying a dead load of 20 kN/m and imposed load of 40 kN/m. The beam is laterally unsupported throughout the span. [14]

Q5) Design a gantry girder supporting an electronically operated crane to the following data : [17]

- a) Capacity of crane = 120 kN
- b) Span between crane rails = 20 m
- c) Self-weight crane girder = 100 kN
- d) Weight of crab, electric motor, Hook etc. = 15 kN
- e) Minimum hook approach = 1.2 m
- f) Wheelbase = 2 m
- g) Span of Gantry 5.5 m

OR

Q6) Determine panel point dead load, imposed load and wind load for the truss as show in Figure 1. Assume design wind pressure as 1200 N/m², use A.C. Sheet and the C/C spacing of truss is 6 m. Assume self-weight of purlin as 120 N/m. [17]

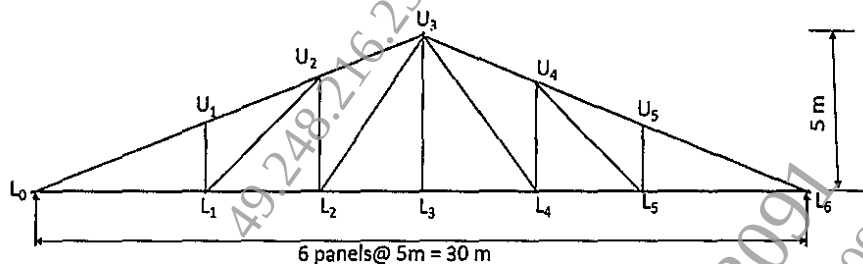


Fig. 1

- Q7) a)** Explain in brief IS provisions for length and spacing of intermittent weld. [4]
- b)** A simply supported welded plate girder of an effective span of 20 m subjected to factored uniformly distributed load 43 kN/m throughout the span including self weight. Assume compression flange laterally supported throughout the span. Design cross section of the plate girder. [14]

OR

Q8) a) Explain in brief flange curtailment of plate girder. **[4]**

b) A Simply supported welded plate girder is designed for the span of 24 m. It is subjected to a shear force of 2300 kN and bending moment of 20700 kNm. A section used for plate girder to carry above load is as given below -

Flanges - 780mm wide and 50 mm thick

Web - 16 mm thick and 2600 mm deep

Design intermittent welded connection between flange and web. Also design end bearing stiffener. **[14]**

